

A. Mohammed Ayub

Machine Learning & AI Engineer

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EXECUTIVE SUMMARY

Fresher AI/ML Engineer with **2 production AI products** shipped during undergrad internships. Hands-on experience building RAG pipelines, LLM powered automation systems, and end to end ML pipelines using Python, FastAPI, and LangChain. Recognized community speaker on Agentic AI and GenAI deployment. Graduating May 2026.

EXPERIENCE

AI/ML Engineer Intern | Radiance Testing Solutions March 2026 – Present | *Pixel Perfect : AI-Powered Visual UI Regression Platform*

- Built an automated Playwright web-scraping layer capturing pixel-accurate screenshots across **2+ device viewports**, eliminating manual screenshot effort and enabling **100% coverage** of responsive breakpoints.
- Integrated Figma Developer API to programmatically export design frames, cutting design to implementation handoff time by **~70%** and removing manual asset downloads entirely.
- Engineered the core visual comparison engine using **Claude Opus 4.7** for human-level reasoning, detecting layout shifts and semantic UI inconsistencies that pixel-diff tools miss — achieving **~90% defect detection accuracy** in internal tests.
- Developed FastAPI backend services orchestrating the export → scrape → compare pipeline, reducing defect reporting turnaround from hours to **under 5 minutes** per run.

Tech: Python, FastAPI, Playwright, Figma Developer API, Claude Opus, Supabase, REST APIs

AI Product Developer Intern | Pibi Technologies 3 months | *Compliance Guard : Automated GST Compliance System*

- Built an end to end GST compliance verification system ingesting GSTR records and running rule based mismatch detection, automating **100% of previously manual** compliance checks.
- Designed real time validation pipelines with multi channel alerting (dashboard + WhatsApp Business API), enabling proactive compliance tracking through automated reminders **5 days and 3 days before filing deadlines**.
- Architected modular backend logic enabling new compliance rules to be added **without core refactors**, keeping the system future-proof as GST regulations evolve.

CRM Knowledge Assistant — RAG Chatbot for Internal CRM

- Built a RAG based chatbot over the company's CRM schema using vector embeddings and retrieval grounded LLM responses, **reducing hallucination rate to near zero** on structural queries and cutting support lookup time by **~60%**.

Tech: Python, FastAPI, REST APIs, LLMs, RAG, Vector Embeddings, WhatsApp Business API, Supabase

PROJECTS

AI Outreach System — B2B Email Automation Platform

- Designed an end to end outreach pipeline automating contact discovery, email verification via third party APIs, and AI personalized email generation for scalable B2B campaigns.
- Integrated Gmail API for automated sending, inbox polling, and reply tracking; built a Streamlit dashboard surfacing real time analytics (opens, replies, and bounces), saving **~8 hours/week** of manual outreach work.

Tech: Python, FastAPI, LLMs (Gemini), Gmail API, Streamlit

ScanBuddy — Multimodal Medical Scan Chatbot

- Developed a multimodal medical scan assistant using **Gemma 1.3B** (via Ollama) with OCR powered scan name extraction from uploaded images/text, supporting **20+ scan types** and contextual scan information retrieval.
- Implemented conversational memory for multi turn interactions, enabling follow up queries without losing prior scan context while providing scan guidance including purpose, procedure, precautions, post-scan care, and warnings.
- Engineered Tamil/Hindi auto translation support to improve accessibility for non English speaking patients.

Tech: Python, Ollama, Gemma 1.3B, OCR, Translation APIs, HTML, CSS, JS

Cyberbullying Text Detection — NLP Classification

- Built a full NLP pipeline (cleaning, tokenization, lemmatization, TF-IDF) over **3 combined datasets**; benchmarked Logistic Regression and SVM with user-level features, achieving **~88% F1-score** vs baseline.

Tech: Python, Scikit-learn, NLTK, Pandas

Telecom Customer Churn Prediction — End to End ML Pipeline

- Engineered **9 domain driven features** (tenure groups, charges per service, contract strength) and eliminated **7 highly correlated features** (>0.8 correlation) to reduce multicollinearity.
- Tuned 4 models (LogReg, DT, RF, XGBoost) with GridSearchCV; built a soft voting ensemble achieving **~86% recall on churners**, outperforming any single model by ~4%.

Tech: Python, Scikit-learn, XGBoost, Pandas, NumPy, Seaborn, Joblib

Time Series Forecasting with Prophet

- Built a weekly sales forecasting pipeline on the Superstore dataset using Facebook Prophet with **log-transformation**, yearly/weekly seasonality, and Category × Segment × Region interaction regressors with growth-capped extrapolation.
- Tuned **changepoint_prior_scale** and seasonality mode via grid search; achieved **MAE: 988, RMSE: 1255, SMAPE: 18.27%** on held-out weekly data.
- Deployed a **Streamlit dashboard** with Plotly for interactive region/segment filtering and forecast vs. actual visualization.

Tech: Python, Prophet, Scikit-learn, Pandas, NumPy, Streamlit, Plotly

TECHNICAL SKILLS

Languages: Python, SQL

ML / AI: Scikit-learn, TensorFlow, Pytorch, XGBoost, Random Forest, SVM, Time Series Forecasting, Feature Engineering, Model Deployment, Hyperparameter Tuning

Generative AI / NLP: LLMs (Claude, GPT, Gemma, Gemini), RAG, Vector Embeddings, LangChain, Prompt Engineering, Hugging Face, Ollama, Multilingual NLP, NLTK, CI/CD

Frameworks / Tools: FastAPI, REST APIs, Streamlit, Playwright, Pandas, NumPy, Matplotlib, Seaborn, Facebook Prophet

Platforms: Git, GitHub, Supabase, Vector DB, Jira, Jupyter, Gmail API, WhatsApp Business API, VS Code

Automation / Testing: Figma Developer API, Web Scraping, Visual Regression Testing, OCR

EDUCATION

B.Tech in Information Technology | K.L.N. College of Engineering, Madurai, Tamil Nadu

CGPA: 7.89 | Expected Graduation: May 2026

ACHIEVEMENTS & COMMUNITY

- **Top 10 Finalist** — Kalasalingam Euphoria 2025 AI/ML Hackathon, Healthcare AI track (competed against 100+ teams).
- **Recognized Speaker** — Madurai AI Community; delivered sessions on Agentic AI & GenAI deployment to **50+ developers, Business Owners and students**.
- **Community Contributor** — Led hands on workshops on RAG pipelines, LLM deployment, and practical GenAI use cases at Madurai AI Community.
- **2 Production AI Products** shipped during undergrad internships Pixel Perfect (visual regression) and Compliance Guard (GST automation).